

Linear Programming Report

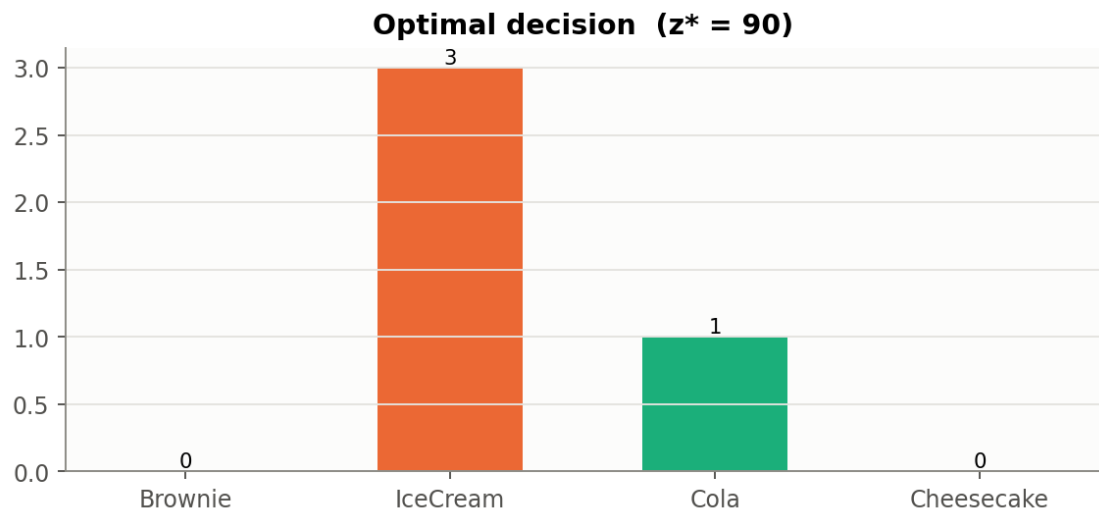
Sensitivity Analysis & Duality

Diet Problem

Meet daily requirements of calories, chocolate, sugar and fat at minimum cost using brownies, ice cream, cola and cheesecake (prices in cents).

Source: Lecture slides 39-47 (dual of a min problem)

Problem type	MIN, 4 variables, 4 constraints
Status	OPTIMAL
Optimal Cost (cents)	90
Dual status	OPTIMAL
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Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
Date	27 September 2026, 15:11



1. Model formulation

MIN $z = 50 \text{ Brownie} + 20 \text{ IceCream} + 30 \text{ Cola} + 80 \text{ Cheesecake}$
 subject to
 $400 \text{ Brownie} + 200 \text{ IceCream} + 150 \text{ Cola} + 500 \text{ Cheesecake} \geq 500$ (Calories)
 $3 \text{ Brownie} + 2 \text{ IceCream} \geq 6$ (Chocolate)
 $2 \text{ Brownie} + 2 \text{ IceCream} + 4 \text{ Cola} + 4 \text{ Cheesecake} \geq 10$ (Sugar)
 $2 \text{ Brownie} + 4 \text{ IceCream} + \text{Cola} + 5 \text{ Cheesecake} \geq 8$ (Fat)
 $\text{Brownie} \geq 0, \text{IceCream} \geq 0, \text{Cola} \geq 0, \text{Cheesecake} \geq 0$

Data table

Constraint	Brownie	IceCream	Cola	Cheesecake	Sense	RHS
Calories	400	200	150	500	$\geq$	500
Chocolate	3	2	0	0	$\geq$	6
Sugar	2	2	4	4	$\geq$	10
Fat	2	4	1	5	$\geq$	8
Objective c	50	20	30	80	min	—

Definitions

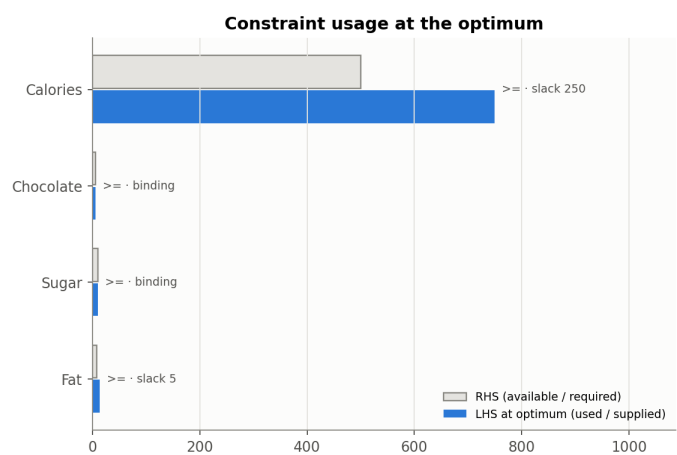
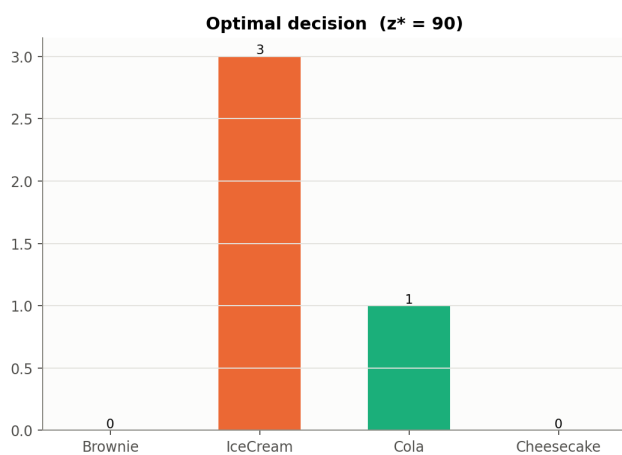
- **Brownie**: brownies eaten
- **IceCream**: scoops of chocolate ice cream
- **Cola**: bottles of cola
- **Cheesecake**: pieces of pineapple cheesecake
- **Calories**: calories
- **Chocolate**: ounces of chocolate
- **Sugar**: ounces of sugar
- **Fat**: ounces of fat

2. Optimal solution

Optimal Cost (cents): **90**

Variable	Description	Optimal value	Obj. coefficient	Contribution
Brownie	brownies eaten	0	50	0
IceCream	scoops of chocolate ice cream	3	20	60
Cola	bottles of cola	1	30	30
Cheesecake	pieces of pineapple cheesecake	0	80	0

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
Calories	750	$\geq$	500	250	no
Chocolate	6	$\geq$	6	0	yes
Sugar	10	$\geq$	10	0	yes
Fat	13	$\geq$	8	5	no



3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
Brownie	0	27.5	50	∞	27.5	22.5	∞
IceCream	3	0	20	18.33	5	15	38.33
Cola	1	0	30	10	30	0	40
Cheesecake	0	50	80	∞	50	30	∞

Right-hand-side ranging (constraints)

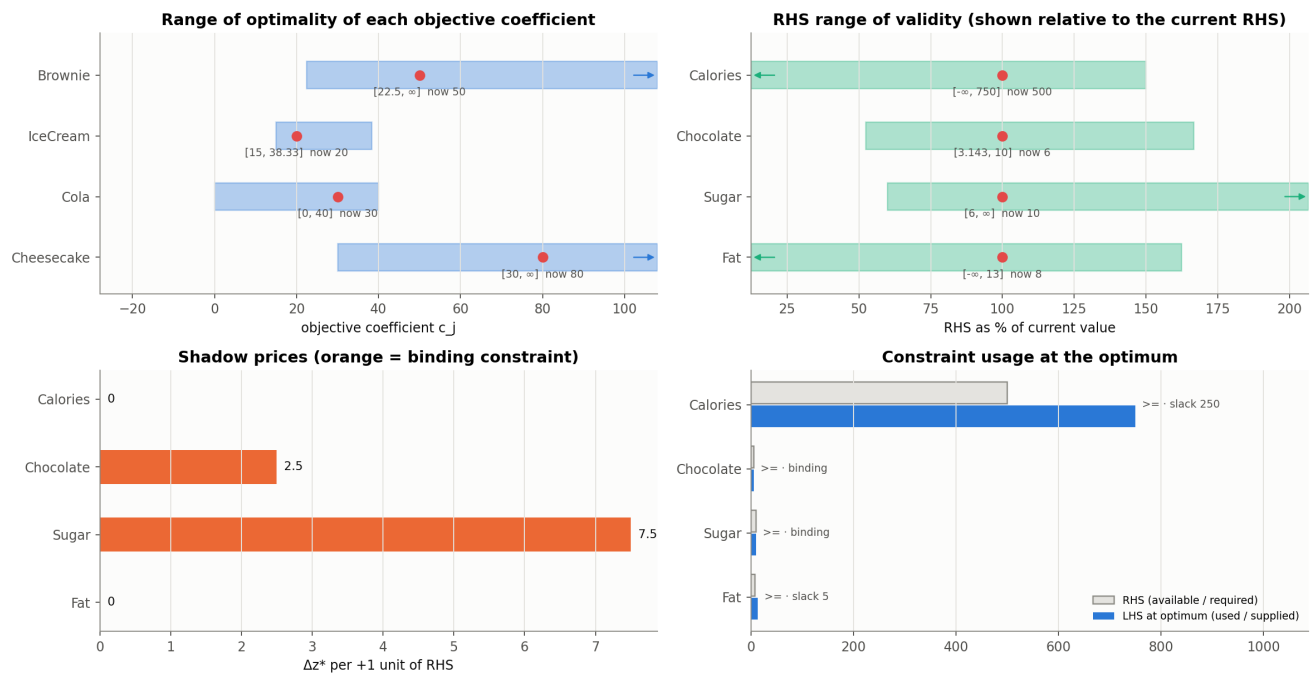
Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
Calories	750	$\geq$	500	250	0	250	∞	$-\infty$	750
Chocolate	6	$\geq$	6	0	2.5	4	2.857	3.143	10
Sugar	10	$\geq$	10	0	7.5	∞	4	6	∞
Fat	13	$\geq$	8	5	0	5	∞	$-\infty$	13

Interpretation

- **Brownie** is not used (value 0). Reduced cost 27.5: its coefficient must decrease by about 27.5 before it becomes attractive.
- **IceCream** = 3. Its objective coefficient (20) may vary in [15, 38.33] without changing the optimal solution (one change at a time).
- **Cola** = 1. Its objective coefficient (30) may vary in [0, 40] without changing the optimal solution (one change at a time).
- **Cheesecake** is not used (value 0). Reduced cost 50: its coefficient must decrease by about 50 before it becomes attractive.
- **Calories** is not binding (slack 250), so its shadow price is 0; the RHS can move within $[-\infty, 750]$ without effect on Cost (cents).
- **Chocolate** is binding. Shadow price 2.5: each extra unit of RHS changes Cost (cents) by 2.5, valid while the RHS stays in [3.143, 10].
- **Sugar** is binding. Shadow price 7.5: each extra unit of RHS changes Cost (cents) by 7.5, valid while the RHS stays in [6, ∞].
- **Fat** is not binding (slack 5), so its shadow price is 0; the RHS can move within $[-\infty, 13]$ without effect on Cost (cents).

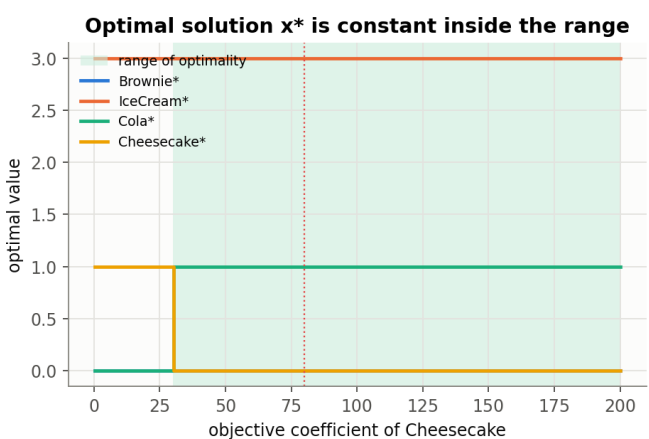
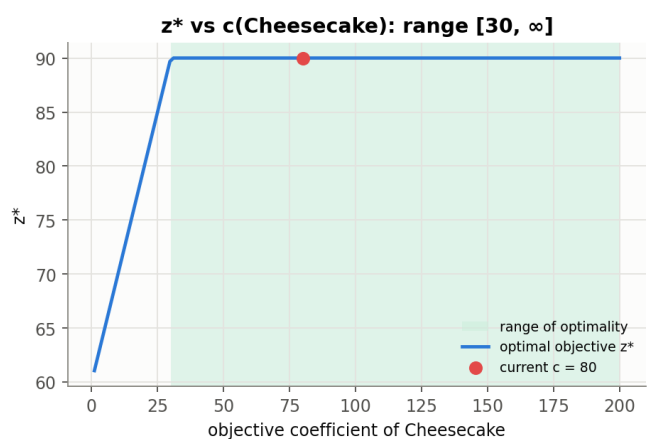
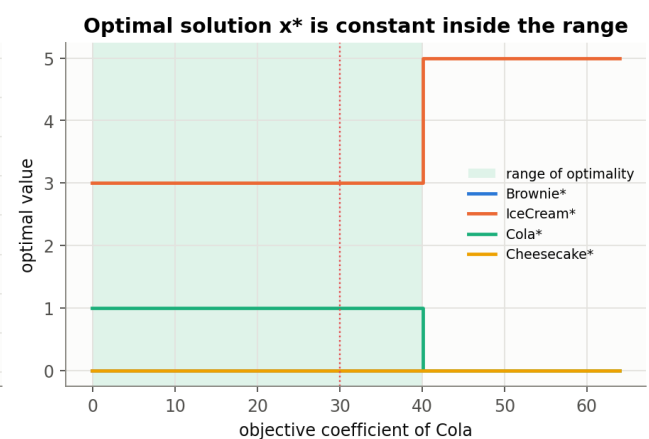
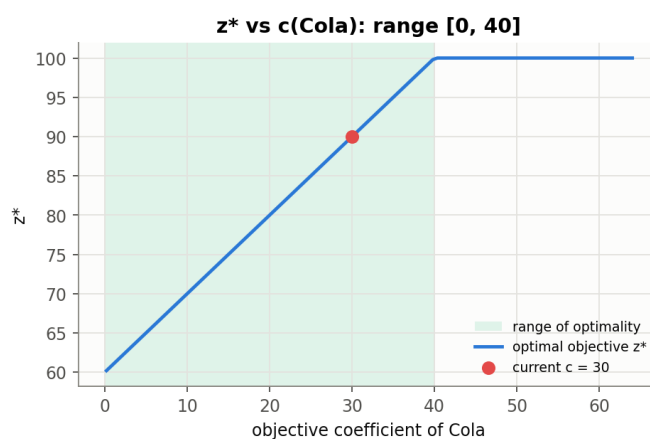
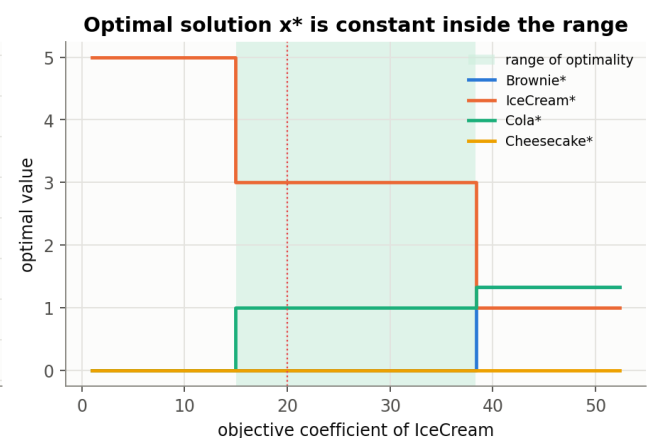
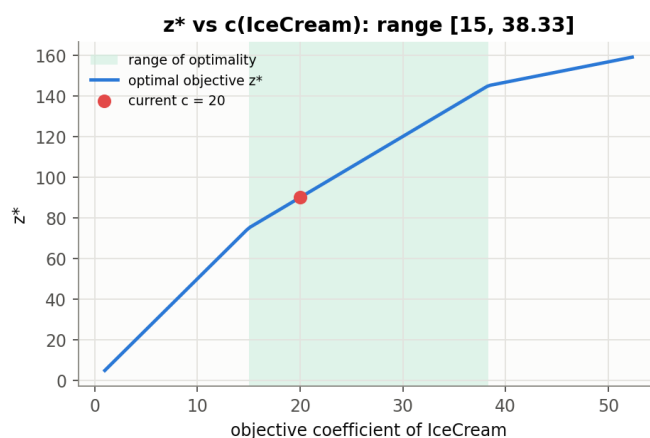
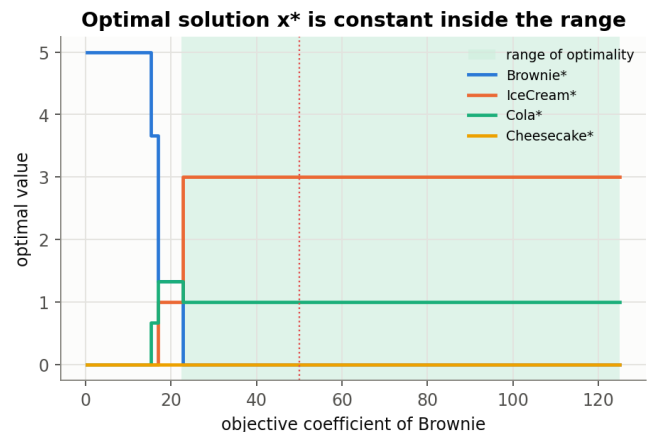
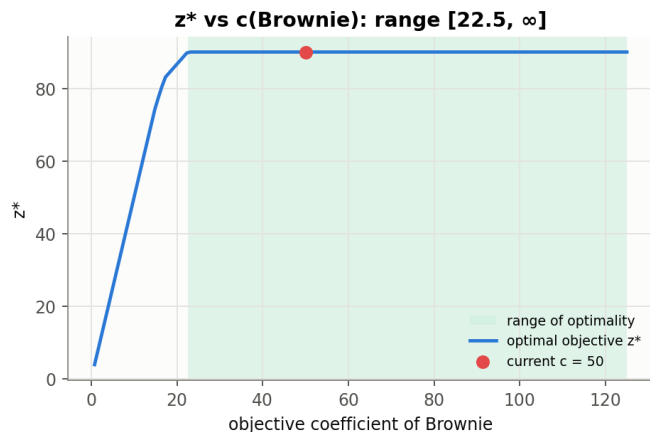
Sensitivity dashboard

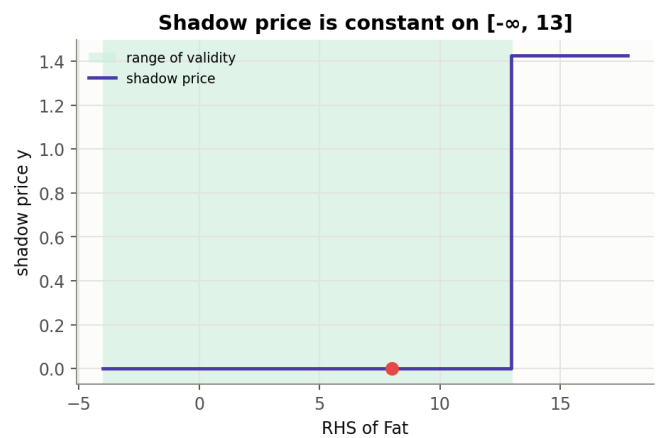
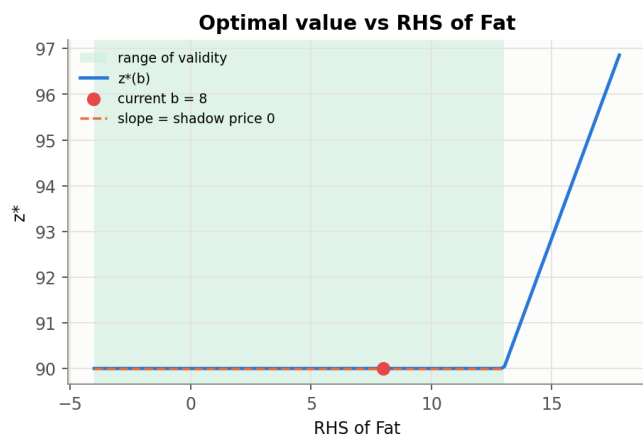
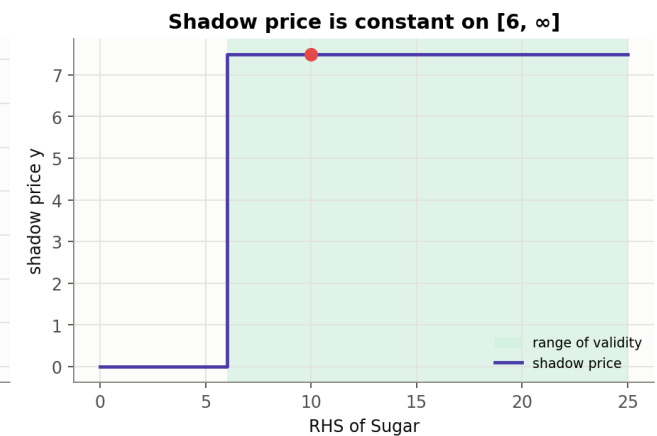
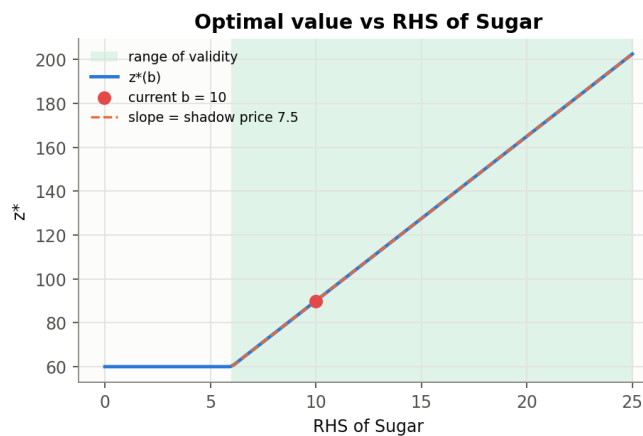
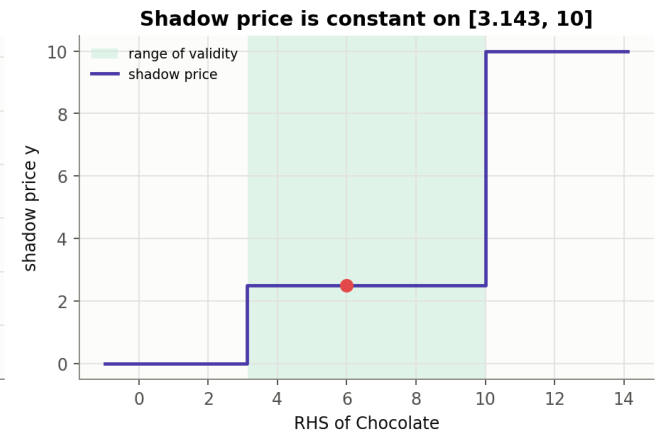
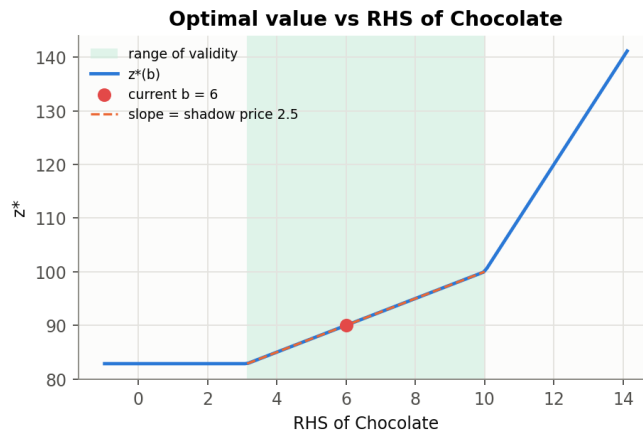
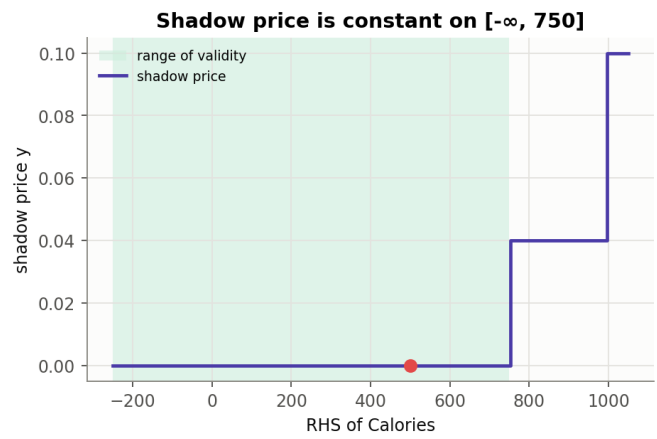
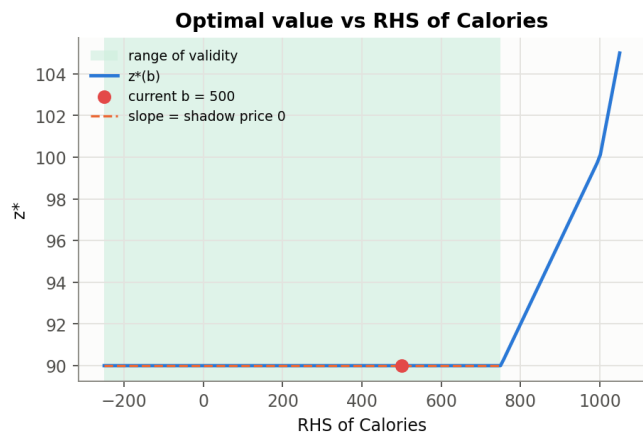
Diet Problem: sensitivity dashboard ($z^* = 90$)



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.





5. What-if scenarios

'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

Scenario	InRange	Predicted	Actual	NewSolution	Note
c(Brownie): 50 → 30	yes	90	90	0, 3, 1, 0	brownie price falls to 30 cents
b(Sugar): 10 → 12	yes	105	105	0, 3, 1.5, 0	sugar requirement rises to 12 oz

6. Duality

Dual formulation

MAX $z = 500 y_{\text{Calories}} + 6 y_{\text{Chocolate}} + 10 y_{\text{Sugar}} + 8 y_{\text{Fat}}$
 subject to
 $400 y_{\text{Calories}} + 3 y_{\text{Chocolate}} + 2 y_{\text{Sugar}} + 2 y_{\text{Fat}} \leq 50$ (Brownie)
 $200 y_{\text{Calories}} + 2 y_{\text{Chocolate}} + 2 y_{\text{Sugar}} + 4 y_{\text{Fat}} \leq 20$ (IceCream)
 $150 y_{\text{Calories}} + 4 y_{\text{Sugar}} + y_{\text{Fat}} \leq 30$ (Cola)
 $500 y_{\text{Calories}} + 4 y_{\text{Sugar}} + 5 y_{\text{Fat}} \leq 80$ (Cheesecake)
 $y_{\text{Calories}} \geq 0, y_{\text{Chocolate}} \geq 0, y_{\text{Sugar}} \geq 0, y_{\text{Fat}} \geq 0$

Dual solution (resource / requirement prices)

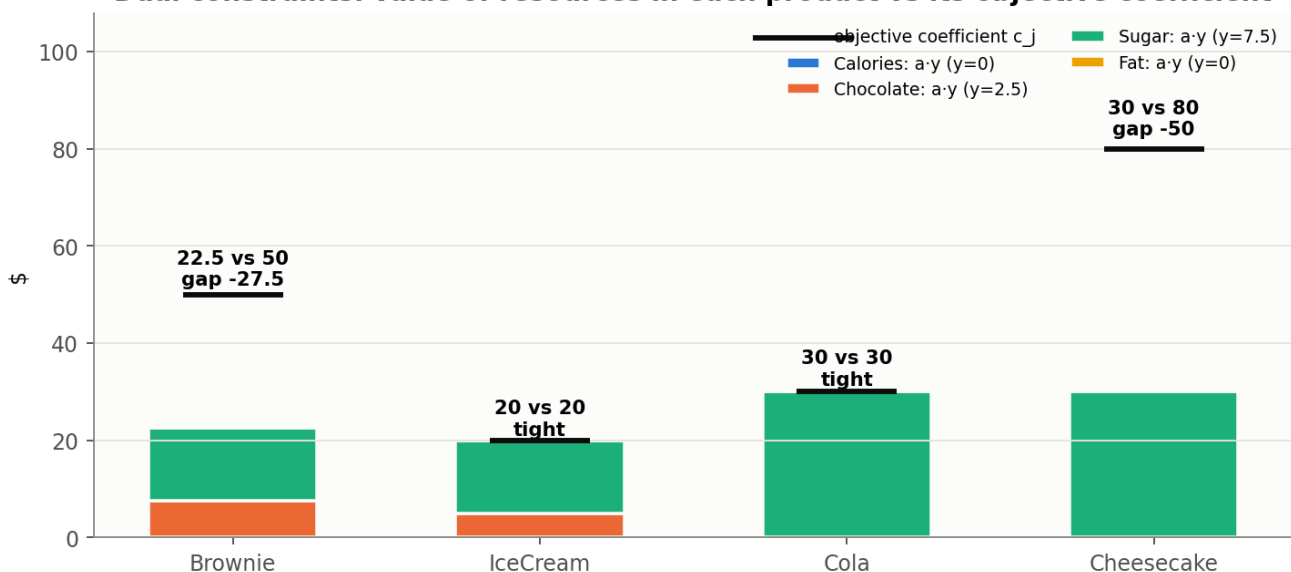
Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
y_{Calories}	Calories	0	0	0
$y_{\text{Chocolate}}$	Chocolate	2.5	2.5	15
y_{Sugar}	Sugar	7.5	7.5	75
y_{Fat}	Fat	0	0	0

Strong duality check: primal $z^* = 90$, dual $w^* = 90$ (equal ✓). Dual of the dual equals the primal: yes.

Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
Calories slack $\times y_{\text{Calories}}$	250	0	0
Chocolate slack $\times y_{\text{Chocolate}}$	0	2.5	0
Sugar slack $\times y_{\text{Sugar}}$	0	7.5	0
Fat slack $\times y_{\text{Fat}}$	5	0	0
Brownie $\times$ surplus of dual constraint Brownie	27.5	0	0
IceCream $\times$ surplus of dual constraint IceCream	-7.105e-15	3	0
Cola $\times$ surplus of dual constraint Cola	7.105e-15	1	0
Cheesecake $\times$ surplus of dual constraint Cheesecake	50	0	0

Dual constraints: value of resources in each product vs its objective coefficient



Primal (min): 4 constraints × 4 variables

50	20	30	80	← objective
400	200	150	500	>= 500
3	2	0	0	>= 6
2	2	4	4	>= 10
2	4	1	5	>= 8
Brownie	IceCream	Cola	Cheesecake	

Dual (max): 4 constraints × 4 variables

500	6	10	8	← objective
400	3	2	2	<= 50
200	2	2	4	<= 20
150	0	4	1	<= 30
500	0	4	5	<= 80
y_Calories	y_Chocolate	y_Sugar	y_Fat	

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_j - c_j$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	1	Cheesecake	a1	524	yes	no	(0, 0, 0, 0)
1	primal	1	IceCream	a4	15	yes	no	(0, 0, 0, 1)
2	primal	1	Brownie	Cheesecake	8.4	yes	no	(0, 1.5, 0, 0.4)
3	primal	1	Cola	Brownie	7	yes	no	(0.3333, 1.833, 0, 0)
4	primal	1	e1	a3	5	yes	no	(0, 1.75, 1, 0)
5	primal	1	Brownie	a2	2.857	yes	no	(0, 1.571, 1.714, 0)
6	primal	1			-0	yes	yes	(1.333, 1, 1.333, 0)
7	primal	2	e4	Brownie	126.7	yes	no	(1.333, 1, 1.333, 0)
8	primal	2			90	yes	yes	(0, 3, 1, 0)

Primal simplex - phase 1, tableau 0
enter Cheesecake, leave a1 (min ratio 1)

	Brownie	IceCream	Cola	Cheesecake	e1	e2	e3	e4	a1	a2	a3	a4	RHS
w	-407	-208	-155	-509	1	1	1	1	0	0	0	0	-524
a1	400	200	150	500	-1	0	0	0	1	0	0	0	500
a2	3	2	0	0	0	-1	0	0	0	1	0	0	6
a3	2	2	4	4	0	0	-1	0	0	0	1	0	10
a4	2	4	1	5	0	0	0	-1	0	0	0	1	8

Primal simplex - phase 2, tableau 8
optimal tableau

	Brownie	IceCream	Cola	Cheesecake	e1	e2	e3	e4	RHS
z	27.5	0	0	50	0	2.5	7.5	0	-90
Cola	-0.25	0	1	1	0	0.25	-0.25	0	1
e4	3.75	0	0	-4	0	-1.75	-0.25	1	5
e1	-138	0	0	-350	1	-62.5	-37.5	0	250
IceCream	1.5	1	0	0	0	-0.5	0	0	3